

Subject: CBX750F wiring problems and solutions.

By: Alan McArdle

I recently bought a 750F, very tidy and low mileage but it had electrical problems. This group has references to voltage regulator, lighting, and tank-gauge problems; they are all parts of one bigger problem that seems common in this bike.

A series of voltage measurements around the wiring loom showed resistance due to poor connections and skinny wire added up to nearly 2 volts lost in some places. The main results are:

- The voltage regulator incorrectly senses low voltage and over-charges the battery. Viking's relay modification can help to fix the effect and stop boiling the battery, but does not fix the cause.
- Lighting is not as bright as one might expect from twin headlights, and another report observed that putting in higher wattage bulbs made no difference (it might actually be worse!).
- Tank gauge and voltmeter readings are hopelessly inaccurate mainly due to common-mode voltages on the negative line.
- Indicator lights are dim

The causes are mainly due to poor contact in the loom connectors, under-sized wiring for the loads (it may have been okay for the earlier single-headlight CB but was not designed for two 60W lamps). My solutions are fairly simple:

- Several contacts in the 6-way spade connector on the fuse panel, ignition switch, and voltage regulator were hot (=bad) and cleaning did not help. Corrosion inside the crimp meant they needed replacing (a similar 6x connector cost \$5.50 from Super-cheap-Auto, I hate to think what Honda would charge).
- The ignition switch can be dismantled (turn the key fully right to enable the rear cap to withdraw after unclipping), the contacts smoothed and polished, with a trace of Vaseline (no ordinary grease) for lubrication. All power goes through the main switch so good electrical contact is vital.
- Negative wiring is badly designed; the battery (–) is grounded at the starter motor then one under-sized common wire connects all services from another ground under the tank. Cleaning these lugs only improved it slightly, so I ran one 4mm² wire from the battery (–) direct to the headlamp negatives (near the dip relay), another from the battery (–) to the instrument circuit (use the voltmeter green) with a tapping on the chassis ground lug, and a link from the ground lug to the v-regulator ground (green).
- The main fuse (30A) beside the starter solenoid had been replaced long ago with a bit of wire! I fitted an enclosed blade-fuse mounting because the fuses are available everywhere, the wires connect to the old fuse terminals and lead out of a slot in the right-hand side of the cover. This is also convenient for isolating power when working on wiring.
- Positive circuits are also overloaded; I separated the headlamp power from the other services to reduce current in the common connectors and ignition switch. A second blade-fuse mount (15A) was fitted beside the main fuse and a 4mm² wire feeds power to the headlamp relay. The old headlamp fuse on the handlebar boss still supplies the relay power, and can be reduced to 5A.

- I modified the tank-gauge to update the 1930's technology thermal regulator to a 1980's LM317 IC, and will describe that separately.

Construction tips:

- Use good-quality lugs, crimp them solidly onto the wire, then solder. No more problems.
- Fit heat-shrink sleeves (many sizes in Dick Smith) on flex or chafe points.
- Bind wiring safely along the wiring loom with cable ties.
- Insulate cut wire-ends with silicon and heat-shrink
- Use the right fuse.

Results:

- Bright headlights! I would avoid fitting higher wattage (100/60W) bulbs, the alternator has capacity but am not sure about the rectifier.
- Easy starting!
- Voltage regulator keeps the battery at the correct voltage (14.2V) over the full rev range whether the lights are on or off.
- Tail, brake, and indicator lamps are brighter, more visible.
- Voltmeter and tank-gauge instruments read correctly.

Modifications

